

CLASSICAL MECHANICS

Unit I and Unit II

Extended Lecture Notes

B.Sc. Physics - Fifth (5^{th}) Semester

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Designed for classroom teaching, self-study and examination revision

Expanded edition with step-by-step derivations and worked examples

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How to Use These Notes

These notes are written as a continuous learning text. The goal is not merely to give formulas but to show why each formula appears and how one idea leads to the next.

The central learning pathwa is

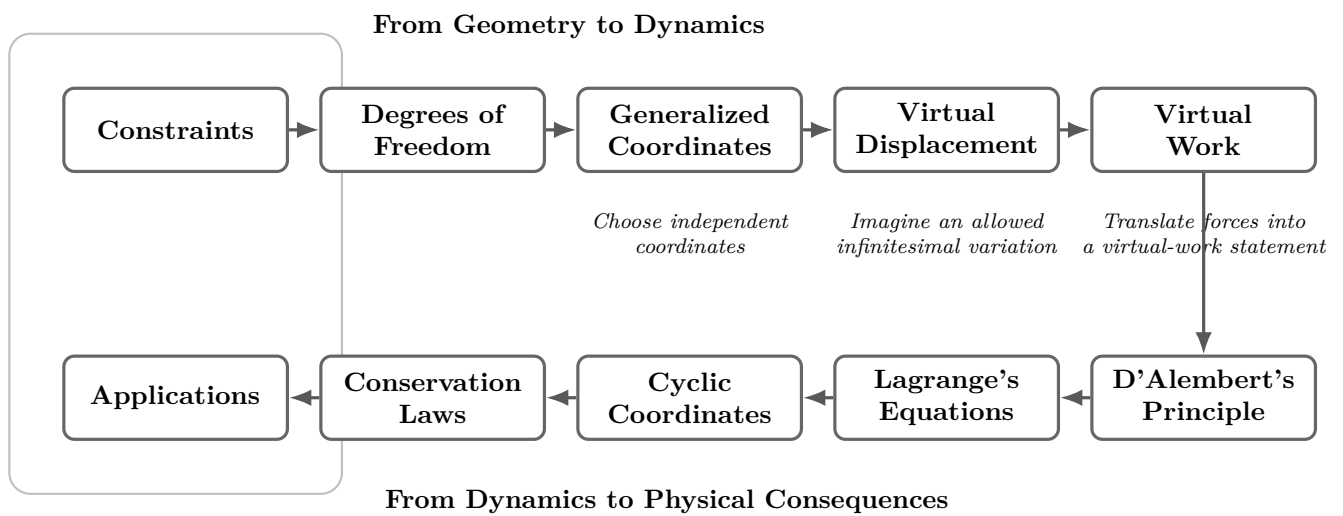


Figure 1: The central learning pathway of analytical mechanics unit I and II. Each step introduces the idea required for the next one, leading naturally from constrained motion to Lagrange's equations and finally to conservation laws and applications.

A useful study method is:

1. Read the physical explanation first.
2. Write down the independent coordinates and constraints.
3. Follow every algebraic step in the derivation.
4. Reproduce the derivation once without looking.
5. Work through the examples before using the formula sheet.

Chapter 1

Complete Outline of Unit I and Unit II

1.1 Unit I - Constrained Motion

1. Meaning and physical need for constrained motion.
2. Constraints: definition and examples.
3. Classification: holonomic/non-holonomic; scleronomous/rheonomous; bilateral/unilateral.
4. Degrees of freedom and configuration space.
5. Constrained systems and forces of constraint.
6. Generalized coordinates and their selection.
7. Transformation equations and generalized notation.
8. Virtual displacement and distinction between δ and d .
9. Virtual work and generalized forces.
10. Principle of virtual work with examples.
11. D'Alembert's principle and its physical interpretation.
12. Detailed derivation of the generalized dynamical equation from D'Alembert's principle.
13. Detailed derivation of Lagrange's equation for conservative systems.

1.2 Unit II - Lagrangian Formalism

1. Lagrangian for conservative systems.
2. Lagrange's equations of motion.
3. Non-conservative systems and generalized forces.
4. Comparison of Newtonian and Lagrangian formulations.
5. Cyclic coordinates and conservation of conjugate momenta.
6. Conservation of energy and the energy integral.
7. Important properties of the kinetic-energy function.
8. Worked examples: free particle, gravity, SHM, pendulum, Atwood machine, central potential and damped oscillator.
9. Problem-solving algorithm, formula sheet, checks, common mistakes and examination questions.

Chapter 2

UNIT I - CONSTRAINED MOTION

2.1 Why Constrained Motion Matters

Newton's laws remain valid for constrained systems. The difficulty is that the coordinates describing the system are no longer all independent.

For a free particle in three dimensions, x, y, z are independent. For a pendulum bob attached to a string of fixed length ℓ ,

$$x^2 + y^2 = \ell^2.$$

Thus x and y cannot be varied independently. Only one independent coordinate is required; the angle θ is a natural choice:

$$x = \ell \sin \theta, \quad y = -\ell \cos \theta.$$

This is the main motivation for analytical mechanics: choose coordinates that describe only the physically possible configurations.

2.2 Constraints: Definition

A constraint is a restriction on the possible configuration or motion of a mechanical system. A general constraint can be written as

$$f(q_1, q_2, \dots, q_n, t) = 0.$$

Examples:

$$\begin{array}{ll} x^2 + y^2 - \ell^2 = 0 & \text{fixed pendulum length,} \\ x^2 + y^2 + z^2 - R^2 = 0 & \text{particle on a sphere.} \end{array}$$

For several independent constraints,

$$f_1 = 0, \quad f_2 = 0, \quad \dots, \quad f_k = 0.$$

2.3 Classification of Constraints

2.3.1 Holonomic Constraints

A constraint is holonomic if it can be expressed as an equation involving coordinates and possibly time:

$$f(q_1, \dots, q_n, t) = 0.$$

Example:

$$x^2 + y^2 - \ell^2 = 0.$$

2.3.2 Non-holonomic Constraints

A restriction that cannot, in general, be integrated into a relation involving only coordinates and time is non-holonomic. A typical differential form is

$$\sum_i a_i(q, t) dq_i + a_t(q, t) dt = 0,$$

when it is non-integrable.

Inequality restrictions such as one-sided contact conditions are also important examples of constraints that are not represented by a single coordinate equation of the holonomic type.

2.3.3 Scleronomous and Rheonomous

A holonomic constraint without explicit time dependence is scleronomous:

$$f(q_1, \dots, q_n) = 0.$$

A constraint containing time explicitly is rheonomous:

$$f(q_1, \dots, q_n, t) = 0.$$

Example: a bead on a stationary circular ring has a scleronomous constraint. If the ring radius is prescribed as $R(t)$, the constraint

$$x^2 + y^2 - R^2(t) = 0$$

is rheonomous.

2.3.4 Bilateral and Unilateral

A bilateral constraint can be expressed as an equality:

$$f(q, t) = 0.$$

A unilateral constraint restricts the system to one side:

$$f(q, t) \geq 0$$

or the corresponding inequality.

Example: a block resting on a horizontal table cannot penetrate the table, but it can lose contact. Hence contact is unilateral.

2.4 Degrees of Freedom

The number of degrees of freedom (DOF) is the number of independent coordinates required to specify the configuration completely.

For N particles in three-dimensional space, there are initially $3N$ coordinates. With k independent holonomic constraints,

$$s = 3N - k.$$

Examples:

- One free particle: $s = 3$.

- One particle on a fixed plane: $s = 2$.
- One particle on a fixed sphere: $s = 2$.
- Pendulum bob in a fixed vertical plane: $s = 2 - 1 = 1$.

A useful question is: “How many independent numbers must I specify at one instant to know the configuration?”

2.5 Configuration and Configuration Space

A configuration is the complete geometrical description of a system at one instant. The configuration space is the abstract space whose coordinates are the independent generalized coordinates.

For a one-dimensional oscillator, the configuration space can be represented by x . For a planar pendulum, it can be represented by θ .

The actual particle moves in physical space; the point representing the configuration moves in configuration space.

2.6 Constrained Systems and Constraint Forces

Constraint forces maintain the restrictions. Examples include tension in a string, normal reaction from a smooth surface, and reaction at a hinge.

For an ideal constraint, the virtual work of the constraint force is zero:

$$\delta W_c = \sum_i \mathbf{F}_{c,i} \cdot \delta \mathbf{r}_i = 0.$$

This does not mean $\mathbf{F}_c = 0$. It means that the force is perpendicular to every admissible virtual displacement.

For a bead on a smooth wire, the reaction is normal to the wire while $\delta \mathbf{r}$ is tangent to the wire:

$$\mathbf{F}_c \cdot \delta \mathbf{r} = 0.$$

2.7 Constrained Motion

Constrained motion is the actual motion of a system when its possible configurations and/or velocities are restricted by one or more constraints. The constraints do not replace Newton’s laws; rather, they reduce the number of independent variables and introduce reaction forces that keep the motion on the allowed path.

For a constrained system the basic logical sequence is

constraint $\longrightarrow$ independent coordinates $\longrightarrow$ allowed motion $\longrightarrow$ equation of motion.

For example, a pendulum bob moving in a vertical plane has Cartesian coordinates (x, y) , but the fixed string length imposes

$$x^2 + y^2 = \ell^2.$$

Thus x and y are not independent. Introduce the angle θ so that

$$x = \ell \sin \theta, \quad y = -\ell \cos \theta.$$

The constraint is then satisfied identically, and the motion is described by the single independent coordinate θ .

The corresponding constraint force is the string tension. It is essential for maintaining the constraint, but for an ideal string its virtual work is zero. Consequently, the tension does not have to be solved explicitly when the Lagrangian method is written in the independent coordinate θ .

For a general constrained system, distinguish:

1. the *constraint*, which specifies what configurations are allowed;
2. the *constraint force*, which enforces that restriction dynamically; and
3. the *constrained motion*, which is the actual trajectory satisfying both the dynamical laws and the constraint.

This distinction becomes especially useful in complicated systems because the Lagrangian formulation allows the constraint to be built into the generalized coordinates from the beginning.

2.8 Generalized Coordinates

Generalized coordinates are any independent variables that uniquely specify the configuration:

$$q_1, q_2, \dots, q_s.$$

They may be lengths, angles or other parameters.

For a pendulum, θ is better than x, y . For planar polar motion, r, ϕ are natural.

Good generalized coordinates:

1. are independent;
2. satisfy geometric constraints automatically when possible;
3. exploit symmetry;
4. make T and V easy to write.

2.9 Transformation Equations

The position of particle i is written as

$$\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, \dots, q_s, t).$$

Differentiating with respect to time,

$$\mathbf{v}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t}.$$

For a time-independent transformation,

$$\mathbf{v}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j.$$

This equation is the bridge between Cartesian mechanics and generalized coordinates.

2.10 Generalized Notation and Important Relations

Suppose a system has s independent degrees of freedom. We denote its generalized coordinates by

$$q_1, q_2, \dots, q_s.$$

The corresponding generalized velocities and accelerations are

$$\dot{q}_j = \frac{dq_j}{dt}, \quad \ddot{q}_j = \frac{d^2q_j}{dt^2}.$$

The position of particle i is expressed as

$$\mathbf{r}_i = \mathbf{r}_i(q_1, q_2, \dots, q_s, t),$$

and hence

$$\mathbf{v}_i = \sum_{j=1}^s \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t}.$$

For a time-independent transformation,

$$\mathbf{v}_i = \sum_{j=1}^s \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j.$$

The virtual displacement at fixed time is

$$\delta \mathbf{r}_i = \sum_{j=1}^s \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j.$$

The generalized force corresponding to q_j is

$$Q_j = \sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j},$$

so that the virtual work is

$$\delta W = \sum_j Q_j \delta q_j.$$

For a conservative potential V ,

$$Q_j = -\frac{\partial V}{\partial q_j}.$$

The generalized momentum conjugate to q_j is

$$p_j = \frac{\partial \mathcal{L}}{\partial \dot{q}_j}.$$

Thus q_j and p_j form a coordinate-momentum pair. If q_j is cyclic, p_j is conserved.

If q_j has dimensions of length, Q_j has dimensions of force. If q_j is an angle, Q_j has the dimensions/physical character of a torque.

2.11 Virtual Displacement

A virtual displacement is an imagined infinitesimal change of configuration at fixed time, consistent with the constraints.

$$\delta \mathbf{r}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j.$$

It is important not to confuse δ with d .

The actual displacement during time dt is

$$d\mathbf{r} = \mathbf{v} dt,$$

whereas $\delta \mathbf{r}$ is an allowed imagined displacement at the same instant.

A useful mental picture is a photograph: freeze time, then imagine moving the system slightly without breaking the constraints.

2.12 Virtual Work

The virtual work done by force $\mathbf{F}_i$ is

$$\delta W_i = \mathbf{F}_i \cdot \delta \mathbf{r}_i.$$

For the whole system,

$$\delta W = \sum_i \mathbf{F}_i \cdot \delta \mathbf{r}_i.$$

Using the transformation for $\delta \mathbf{r}_i$,

$$\begin{aligned} \delta W &= \sum_i \mathbf{F}_i \cdot \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j \\ &= \sum_j \left[\sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right] \delta q_j. \end{aligned}$$

Define the generalized force

$$Q_j = \sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}.$$

Therefore

$$\delta W = \sum_j Q_j \delta q_j.$$

If q_j is a length, Q_j has force dimensions. If q_j is an angle, Q_j has torque dimensions.

2.13 Principle of Virtual Work

For a system in equilibrium subject to ideal constraints,

$$\delta W = 0$$

for every admissible virtual displacement.

Thus

$$\sum_j Q_j \delta q_j = 0.$$

Since the independent δq_j are arbitrary,

$$Q_j = 0$$

for each independent coordinate.

2.13.1 Example: Block on a Smooth Horizontal Surface

Let a block be acted upon by a horizontal force F and vertical normal reaction N and weight mg . An admissible virtual displacement is horizontal:

$$\delta \mathbf{r} = \delta x \hat{\mathbf{x}}.$$

The vertical forces have zero virtual work:

$$N \hat{\mathbf{y}} \cdot \delta x \hat{\mathbf{x}} = 0, \quad (-mg \hat{\mathbf{y}}) \cdot \delta x \hat{\mathbf{x}} = 0.$$

The applied horizontal force gives

$$\delta W = F \delta x.$$

For equilibrium and arbitrary δx ,

$$F = 0.$$

The point is that the normal reaction never had to be solved.

2.13.2 Example: Simple Pendulum in Equilibrium

For a bob at angle θ , the tension is radial and the virtual displacement is tangential:

$$\mathbf{T} \cdot \delta \mathbf{r} = 0.$$

The gravitational force gives

$$\delta W = -mg\ell \sin \theta \delta \theta.$$

Hence equilibrium requires

$$-mg\ell \sin \theta = 0,$$

so

$$\theta = 0, \pi$$

within the usual idealized model. The lower position is stable and the upper position is unstable.

2.14 D'Alembert's Principle

Newton's second law for particle i is

$$\mathbf{F}_i = m_i \mathbf{a}_i.$$

Rearrange:

$$\mathbf{F}_i - m_i \mathbf{a}_i = 0.$$

Multiply by an admissible virtual displacement and sum:

$$\sum_i (\mathbf{F}_i - m_i \mathbf{a}_i) \cdot \delta \mathbf{r}_i = 0.$$

This is D'Alembert's principle.

It can be interpreted by introducing the formal inertial force

$$\mathbf{F}_{\text{in},i} = -m_i \mathbf{a}_i.$$

Then

$$\sum_i (\mathbf{F}_i + \mathbf{F}_{\text{in},i}) \cdot \delta \mathbf{r}_i = 0.$$

The principle is therefore a dynamic version of virtual work.

2.15 Detailed Derivation: D'Alembert's Principle to Lagrange's Equation

This derivation is central to Unit I and Unit II. Every step is shown so that the final Euler-Lagrange equation is understood rather than memorized.

2.15.1 Step 1: Start from D'Alembert's Principle

$$\sum_i (\mathbf{F}_i - m_i \mathbf{a}_i) \cdot \delta \mathbf{r}_i = 0.$$

2.15.2 Step 2: Insert the Virtual Displacement

For independent generalized coordinates,

$$\delta \mathbf{r}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j.$$

Therefore

$$\sum_i (\mathbf{F}_i - m_i \mathbf{a}_i) \cdot \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j = 0.$$

Interchange the sums:

$$\sum_j \left[\sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} - \sum_i m_i \mathbf{a}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right] \delta q_j = 0.$$

2.15.3 Step 3: Define the Generalized Force

Define

$$Q_j = \sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}.$$

Then

$$\sum_j \left[Q_j - \sum_i m_i \mathbf{a}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right] \delta q_j = 0.$$

2.15.4 Step 4: Introduce the Kinetic Energy

Define

$$T = \frac{1}{2} \sum_i m_i v_i^2.$$

For a time-independent coordinate transformation,

$$\mathbf{v}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j.$$

Hence

$$\frac{\partial \mathbf{v}_i}{\partial \dot{q}_j} = \frac{\partial \mathbf{r}_i}{\partial q_j}.$$

Differentiate T with respect to $\dot{q}_j$:

$$\begin{aligned}\frac{\partial T}{\partial \dot{q}_j} &= \frac{1}{2} \sum_i m_i \frac{\partial}{\partial \dot{q}_j} (\mathbf{v}_i \cdot \mathbf{v}_i) \\ &= \sum_i m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{v}_i}{\partial \dot{q}_j} \\ &= \sum_i m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}.\end{aligned}$$

Thus

$$\frac{\partial T}{\partial \dot{q}_j} = \sum_i m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}.$$

2.15.5 Step 5: Differentiate with Respect to Time

Take the total time derivative:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) = \sum_i m_i \mathbf{a}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} + \sum_i m_i \mathbf{v}_i \cdot \frac{d}{dt} \left(\frac{\partial \mathbf{r}_i}{\partial q_j} \right).$$

For a time-independent transformation,

$$\frac{d}{dt} \left(\frac{\partial \mathbf{r}_i}{\partial q_j} \right) = \frac{\partial \mathbf{v}_i}{\partial q_j}.$$

Therefore

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) = \sum_i m_i \mathbf{a}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} + \sum_i m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{v}_i}{\partial q_j}.$$

But

$$\frac{\partial T}{\partial q_j} = \sum_i m_i \mathbf{v}_i \cdot \frac{\partial \mathbf{v}_i}{\partial q_j}.$$

Hence

$$\sum_i m_i \mathbf{a}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} = \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j}.$$

This identity is the mathematical bridge between Newton's acceleration term and the kinetic-energy function.

2.15.6 Step 6: Substitute into D'Alembert's Equation

We had

$$\sum_j \left[Q_j - \sum_i m_i \mathbf{a}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j} \right] \delta q_j = 0.$$

Substitute the identity:

$$\sum_j \left[Q_j - \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) + \frac{\partial T}{\partial q_j} \right] \delta q_j = 0.$$

Rearrange:

$$\sum_j \left[\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} - Q_j \right] \delta q_j = 0.$$

2.15.7 Step 7: Use Independence of the Virtual Variations

Because the generalized coordinates are independent, the virtual variations

$$\delta q_1, \delta q_2, \dots, \delta q_s$$

are independent. Therefore each coefficient must vanish:

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j.$$

This is the generalized dynamical equation.

2.15.8 Step 8: Conservative Forces

For a conservative system,

$$Q_j = -\frac{\partial V}{\partial q_j}.$$

Therefore

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = 0.$$

Define

$$\mathcal{L} = T - V.$$

Since

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_j} = \frac{\partial T}{\partial \dot{q}_j},$$

and

$$\frac{\partial \mathcal{L}}{\partial q_j} = \frac{\partial T}{\partial q_j} - \frac{\partial V}{\partial q_j},$$

we obtain

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0.$$

This is Lagrange's equation.

Chapter 3

UNIT II - LAGRANGIAN FORMALISM

3.1 The Lagrangian

For a conservative mechanical system,

$$\mathcal{L} = T - V.$$

The Lagrangian is not usually the total mechanical energy. For ordinary conservative systems,

$$E = T + V, \quad \mathcal{L} = T - V.$$

The Lagrangian is a scalar function of

$$\mathcal{L} = \mathcal{L}(q_1, \dots, q_s, \dot{q}_1, \dots, \dot{q}_s, t).$$

3.2 Lagrange's Equation

For every independent generalized coordinate q_j ,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0.$$

The safest practical method is:

Choose $q_j \rightarrow$ find $T \rightarrow$ find $V \rightarrow \mathcal{L} = T - V \rightarrow$ differentiate.

3.2.1 Lagrange's Equation for a Conservative System

For a conservative system the potential energy depends on the generalized coordinates:

$$V = V(q_1, q_2, \dots, q_s),$$

and the generalized conservative force is

$$Q_j = -\frac{\partial V}{\partial q_j}. \tag{1}$$

From the generalized dynamical equation obtained from D'Alembert's principle,

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j. \tag{2}$$

Substituting Eq. (1) into Eq. (2),

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial V}{\partial q_j} = 0. \quad (3)$$

Now define the Lagrangian

$$\mathcal{L} = T - V. \quad (4)$$

Since V is independent of generalized velocities,

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_j} = \frac{\partial T}{\partial \dot{q}_j}, \quad (5)$$

and

$$\frac{\partial \mathcal{L}}{\partial q_j} = \frac{\partial T}{\partial q_j} - \frac{\partial V}{\partial q_j}. \quad (6)$$

Substitution of Eqs. (5) and (6) into Eq. (3) gives

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0.$$

Hence, for every independent generalized coordinate,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0.$$

This is Lagrange's equation of motion for a conservative system. The important point is that the potential-energy relation converts the generalized-force equation into the compact Euler-Lagrange form.

3.3 Lagrange's Equation for a Non-Conservative System

When non-conservative forces are present, start from

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = Q_j. \quad (1)$$

Separate the generalized force into conservative and non-conservative parts:

$$Q_j = Q_j^{(c)} + Q_j^{(nc)}.$$

For the conservative part,

$$Q_j^{(c)} = -\frac{\partial V}{\partial q_j}.$$

Thus

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} = -\frac{\partial V}{\partial q_j} + Q_j^{(nc)}.$$

Define

$$\mathcal{L} = T - V.$$

Then

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_j} = \frac{\partial T}{\partial \dot{q}_j}, \quad \frac{\partial \mathcal{L}}{\partial q_j} = \frac{\partial T}{\partial q_j} - \frac{\partial V}{\partial q_j}.$$

Therefore

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_j^{(nc)}. \quad (2)$$

Equation (2) is the Lagrange equation for a system containing non-conservative generalized forces. For a viscously damped oscillator,

$$Q_x^{(nc)} = -c\dot{x},$$

so the damping force is simply placed on the right-hand side.

3.3.1 Special Case: Generalized Potential Depending on Velocity

A force may sometimes be represented by a generalized potential $U(q, \dot{q}, t)$ such that

$$Q_j = -\frac{\partial U}{\partial q_j} + \frac{d}{dt} \left(\frac{\partial U}{\partial \dot{q}_j} \right). \quad (3)$$

Substitution into Eq. (1) gives

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} &= -\frac{\partial U}{\partial q_j} + \frac{d}{dt} \left(\frac{\partial U}{\partial \dot{q}_j} \right), \\ \frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} - \frac{\partial U}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} + \frac{\partial U}{\partial q_j} &= 0. \end{aligned}$$

If we define

$$\mathcal{L} = T - U,$$

then

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_j} = \frac{\partial T}{\partial \dot{q}_j} - \frac{\partial U}{\partial \dot{q}_j}, \quad \frac{\partial \mathcal{L}}{\partial q_j} = \frac{\partial T}{\partial q_j} - \frac{\partial U}{\partial q_j},$$

and hence

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0.$$

For the ordinary conservative case $U = V(q)$, the velocity derivative of U vanishes and the familiar $\mathcal{L} = T - V$ is recovered.

Important: ordinary viscous damping is handled here by the generalized-force form $Q_j^{(\text{nc})}$.

3.4 Comparison of Newtonian and Lagrangian Formulations

Newtonian and Lagrangian mechanics describe the same classical dynamics, but they organize the calculation in different ways. The following comparison incorporates the important points emphasized in the accompanying handwritten classroom notes.

Aspect	Newtonian formulation	Lagrangian formulation
Basic principle	Based directly on Newton's laws of motion.	Based on the Lagrangian formulation of dynamics; for a conservative system $\mathcal{L} = T - V$.
Fundamental equation of motion	$\mathbf{F} = m\mathbf{a}$; in one dimension $F_x = m\ddot{x}$.	$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = Q_j^{(\text{nc})}$; for conservative systems the right side is zero.
Coordinates	Usually Cartesian coordinates (x, y, z) .	Generalized coordinates $(q_1, q_2, \dots, q_s)$ chosen to describe independent motion.
Forces	Individual forces and their components are written explicitly.	Kinetic and potential energies are emphasized; non-conservative forces enter as generalized forces.
Constraints	Constraint forces often have to be introduced and calculated explicitly.	Ideal constraints can often be incorporated through the generalized coordinates, so their reaction forces need not appear explicitly.
Multiple-particle systems	Can become lengthy because every particle and force must be tracked.	Often considerably simpler for coupled and constrained systems because one works with independent coordinates and one scalar Lagrangian.

Applications	Projectile motion, direct force analysis, simple force problems and elementary dynamics.	Pendulums, Atwood machines, coupled oscillators, central-force motion, robotics, celestial mechanics and many-particle systems.
Nature of equations	Vector equations in physical space.	One scalar equation for each independent generalized coordinate.

For a free particle moving along x , Newtonian mechanics gives

$$F_x = m\ddot{x}.$$

In the Lagrangian formulation,

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2,$$

so

$$\frac{\partial \mathcal{L}}{\partial x} = 0, \quad \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}.$$

Lagrange's equation gives

$$\frac{d}{dt}(m\dot{x}) = 0 \implies m\ddot{x} = 0.$$

Thus the two formulations give the same physical result; the advantage of the Lagrangian formulation is often that the coordinates and constraints can be chosen so that the calculation is much shorter.

3.5 Cyclic Coordinates and Conservation of Momentum

A generalized coordinate q_k is called *cyclic* (or ignorable) if it does not occur explicitly in the Lagrangian. It may still occur through its velocity $\dot{q}_k$:

$$\frac{\partial \mathcal{L}}{\partial q_k} = 0.$$

Start from Lagrange's equation,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k} \right) - \frac{\partial \mathcal{L}}{\partial q_k} = 0.$$

For a cyclic coordinate,

$$\frac{\partial \mathcal{L}}{\partial q_k} = 0,$$

therefore

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k} \right) = 0.$$

Define the conjugate momentum

$$p_k \equiv \frac{\partial \mathcal{L}}{\partial \dot{q}_k}.$$

Then

$$\frac{dp_k}{dt} = 0,$$

which integrates to

$$p_k = \text{constant}.$$

This is the mathematical reason that a cyclic coordinate gives a conservation law: absence of q_k from $\mathcal{L}$ makes its conjugate momentum time-independent.

3.5.1 How Linear Momentum is Conserved Mathematically

Consider a free particle moving along x :

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2.$$

The coordinate x is absent explicitly, so

$$\frac{\partial \mathcal{L}}{\partial x} = 0.$$

Its conjugate momentum is

$$p_x = \frac{\partial \mathcal{L}}{\partial \dot{x}} = \frac{\partial}{\partial \dot{x}} \left(\frac{1}{2}m\dot{x}^2 \right) = m\dot{x}.$$

Lagrange's equation gives

$$\frac{dp_x}{dt} = 0.$$

Hence

$$\frac{d}{dt}(m\dot{x}) = 0 \implies m\ddot{x} = 0,$$

and therefore

$$p_x = m\dot{x} = \text{constant}.$$

So the ordinary statement “linear momentum is conserved” follows directly from the cyclic nature of the translational coordinate.

For free motion in three dimensions,

$$\mathcal{L} = \frac{1}{2}m(\dot{x}^2 + \dot{y}^2 + \dot{z}^2).$$

All three coordinates are cyclic, giving

$$p_x = m\dot{x}, \quad p_y = m\dot{y}, \quad p_z = m\dot{z},$$

all constant. Therefore

$$\mathbf{p} = m\mathbf{v} = \text{constant}.$$

3.5.2 Angular Momentum from a Cyclic Coordinate

For a particle in a central potential $V(r)$, use polar coordinates (r, ϕ) :

$$\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) - V(r).$$

The angle ϕ does not occur explicitly:

$$\frac{\partial \mathcal{L}}{\partial \phi} = 0.$$

Thus ϕ is cyclic and

$$p_\phi = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = mr^2\dot{\phi} = \text{constant}.$$

Since

$$L_z = mr^2\dot{\phi},$$

we obtain

$$L_z = \text{constant}.$$

Thus conservation of angular momentum is obtained mathematically from the absence of ϕ from the Lagrangian.

3.5.3 The Conservation Chain to Remember

$$\text{cyclic } q_k \implies \frac{\partial \mathcal{L}}{\partial q_k} = 0 \implies \frac{dp_k}{dt} = 0 \implies p_k = \text{constant}.$$

Therefore a cyclic translational coordinate gives conserved linear momentum, while a cyclic angular coordinate gives conserved angular momentum.

Important: the cyclic coordinate itself is not necessarily constant; its conjugate momentum is constant.

3.6 Conservation of Energy

Define

$$E = \sum_j \dot{q}_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} - \mathcal{L}.$$

Differentiate:

$$\frac{dE}{dt} = \sum_j \ddot{q}_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} + \sum_j \dot{q}_j \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{d\mathcal{L}}{dt}.$$

Using

$$\frac{d\mathcal{L}}{dt} = \sum_j \frac{\partial \mathcal{L}}{\partial q_j} \dot{q}_j + \sum_j \frac{\partial \mathcal{L}}{\partial \dot{q}_j} \ddot{q}_j + \frac{\partial \mathcal{L}}{\partial t},$$

the $\ddot{q}_j$ terms cancel. Hence

$$\frac{dE}{dt} = \sum_j \dot{q}_j \left[\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} \right] - \frac{\partial \mathcal{L}}{\partial t}.$$

By Lagrange's equations,

$$\frac{dE}{dt} = -\frac{\partial \mathcal{L}}{\partial t}.$$

Therefore, if

$$\frac{\partial \mathcal{L}}{\partial t} = 0,$$

then

$$E = \text{constant}.$$

For the standard form $\mathcal{L} = T - V$, with T quadratic in velocities and V independent of velocities,

$$E = T + V.$$

3.7 Properties of the Kinetic-Energy Function

For a time-independent coordinate transformation, the kinetic energy is quadratic in generalized velocities:

$$T = \frac{1}{2} \sum_j \sum_k a_{jk}(q) \dot{q}_j \dot{q}_k.$$

Since T is homogeneous of degree two in $\dot{q}_j$, Euler's theorem gives

$$\sum_j \dot{q}_j \frac{\partial T}{\partial \dot{q}_j} = 2T.$$

This identity is very useful in energy proofs.

If the transformation depends explicitly on time,

$$\mathbf{v}_i = \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t},$$

and T can contain terms linear in velocities as well as explicit time dependence.

Chapter 4

Worked Examples - Step by Step

4.1 Example 1: Free Particle

Let a particle of mass m move freely along x .

Step 1: Coordinate

$$q = x.$$

Step 2: Kinetic energy

$$T = \frac{1}{2}m\dot{x}^2.$$

Step 3: Potential energy

Take $V = 0$.

Step 4: Lagrangian

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2.$$

Step 5: Derivatives

$$\frac{\partial \mathcal{L}}{\partial x} = 0, \quad \frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}.$$

Step 6: Lagrange equation

$$\frac{d}{dt}(m\dot{x}) = 0.$$

Therefore

$$m\ddot{x} = 0, \quad \ddot{x} = 0.$$

Thus $\dot{x}$ is constant and the particle moves uniformly.

4.2 Example 2: Particle Falling Under Gravity

Consider a particle of mass m moving vertically under gravity. Choose upward as positive and let y be measured from an arbitrary reference level.

Step 1: Coordinate and velocity

There is one degree of freedom:

$$q = y, \quad \dot{q} = \dot{y}.$$

Step 2: Kinetic energy

The particle has speed $|\dot{y}|$, so

$$T = \frac{1}{2}m\dot{y}^2.$$

Step 3: Potential energy

Near the Earth's surface,

$$F_y = -mg.$$

For a conservative force,

$$F_y = -\frac{dV}{dy},$$

so

$$-mg = -\frac{dV}{dy} \implies \frac{dV}{dy} = mg.$$

Integrating,

$$V = mgy + C.$$

Choose the zero of potential energy so that $C = 0$:

$$V = mgy.$$

Step 4: Lagrangian

$$\mathcal{L} = T - V = \frac{1}{2}m\dot{y}^2 - mgy.$$

Step 5: Equation of motion

$$\frac{\partial \mathcal{L}}{\partial \dot{y}} = m\dot{y}, \quad \frac{\partial \mathcal{L}}{\partial y} = -mg.$$

Thus

$$\frac{d}{dt}(m\dot{y}) - (-mg) = 0,$$

which gives

$$m\ddot{y} + mg = 0 \implies \ddot{y} = -g.$$

The negative sign follows from choosing upward as the positive direction.

4.3 Example 3: Simple Harmonic Oscillator

Consider a mass m attached to a light spring of force constant k , with the equilibrium position chosen as $x = 0$.

Step 1: Coordinate and velocity

There is one degree of freedom:

$$q = x, \quad \dot{q} = \dot{x}.$$

Step 2: Kinetic energy

The mass has speed $|\dot{x}|$, hence

$$T = \frac{1}{2}m\dot{x}^2.$$

Step 3: Potential energy of the spring

Hooke's law is

$$F_x = -kx.$$

Since the force is conservative,

$$F_x = -\frac{dV}{dx},$$

so

$$-kx = -\frac{dV}{dx} \implies \frac{dV}{dx} = kx.$$

Integrating,

$$V = \frac{1}{2}kx^2 + C.$$

Taking $V = 0$ at equilibrium gives $C = 0$ and hence

$$V = \frac{1}{2}kx^2.$$

Step 4: Lagrangian

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2.$$

Step 5: Equation of motion

$$\frac{\partial \mathcal{L}}{\partial \dot{x}} = m\dot{x}, \quad \frac{\partial \mathcal{L}}{\partial x} = -kx.$$

Therefore

$$\frac{d}{dt}(m\dot{x}) + kx = 0,$$

so

$$m\ddot{x} + kx = 0.$$

Writing

$$\omega^2 = \frac{k}{m},$$

we obtain

$$\ddot{x} + \omega^2 x = 0, \quad \omega = \sqrt{\frac{k}{m}}.$$

The kinetic energy comes from the translational motion of the mass, while the potential energy comes from integrating the spring's restoring force.

4.4 Example 4: Simple Pendulum

Consider a bob of mass m attached to a massless string of length ℓ . Use θ as the generalized coordinate.

The speed is

$$v = \ell \dot{\theta}.$$

Therefore

$$T = \frac{1}{2} m \ell^2 \dot{\theta}^2.$$

Choose the lowest point as zero potential:

$$V = mg\ell(1 - \cos \theta).$$

Thus

$$\mathcal{L} = \frac{1}{2} m \ell^2 \dot{\theta}^2 - mg\ell(1 - \cos \theta).$$

Now

$$\frac{\partial \mathcal{L}}{\partial \theta} = -mg\ell \sin \theta,$$

and

$$\frac{\partial \mathcal{L}}{\partial \dot{\theta}} = m \ell^2 \dot{\theta}.$$

Differentiate:

$$\frac{d}{dt} (m \ell^2 \dot{\theta}) = m \ell^2 \ddot{\theta}.$$

Lagrange equation:

$$m \ell^2 \ddot{\theta} + mg\ell \sin \theta = 0.$$

Divide by $m \ell^2$:

$$\ddot{\theta} + \frac{g}{\ell} \sin \theta = 0.$$

For small oscillations,

$$\sin \theta \simeq \theta,$$

so

$$\ddot{\theta} + \frac{g}{\ell} \theta = 0.$$

Hence

$$\omega = \sqrt{\frac{g}{\ell}}.$$

The approximation is valid only when $|\theta| \ll 1$ radian.

4.5 Example 5: Atwood Machine

Two masses m_1 and m_2 are connected by an inextensible string over a frictionless pulley. Let x be the downward displacement of m_1 .

Because the string length is fixed, if m_1 moves downward by x , m_2 moves upward by x . Thus there is only one independent coordinate.

Both masses have speed magnitude $\dot{x}$:

$$T = \frac{1}{2}m_1\dot{x}^2 + \frac{1}{2}m_2\dot{x}^2 = \frac{1}{2}(m_1 + m_2)\dot{x}^2.$$

Take upward as positive for potential energy. If m_1 moves downward by x , its height decreases by x ; m_2 rises by x . Therefore, apart from an irrelevant constant,

$$V = m_2gx - m_1gx = (m_2 - m_1)gx.$$

Thus

$$\mathcal{L} = \frac{1}{2}(m_1 + m_2)\dot{x}^2 - (m_2 - m_1)gx.$$

Now

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial x} &= -(m_2 - m_1)g, \\ \frac{\partial \mathcal{L}}{\partial \dot{x}} &= (m_1 + m_2)\dot{x}.\end{aligned}$$

Lagrange equation:

$$(m_1 + m_2)\ddot{x} + (m_2 - m_1)g = 0.$$

Therefore

$$\ddot{x} = \frac{(m_1 - m_2)g}{m_1 + m_2}.$$

Notice that the tension does not appear. It is an internal constraint force eliminated by the one-coordinate description.

4.6 Example 6: Particle in a Central Potential

Use plane polar coordinates (r, ϕ) .

The kinetic energy is

$$T = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2),$$

and

$$V = V(r).$$

Thus

$$\mathcal{L} = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\phi}^2) - V(r).$$

Angular equation

Since ϕ is absent,

$$\frac{\partial \mathcal{L}}{\partial \phi} = 0.$$

Therefore

$$\frac{d}{dt} (mr^2 \dot{\phi}) = 0,$$

so

$$mr^2 \dot{\phi} = \text{constant}.$$

This is conservation of angular momentum.

Radial equation

We have

$$\frac{\partial \mathcal{L}}{\partial r} = mr\dot{\phi}^2 - \frac{dV}{dr},$$

and

$$\frac{\partial \mathcal{L}}{\partial \dot{r}} = m\dot{r}.$$

Hence

$$m\ddot{r} - \left(mr\dot{\phi}^2 - \frac{dV}{dr} \right) = 0.$$

Therefore

$$m\ddot{r} - mr\dot{\phi}^2 = -\frac{dV}{dr}.$$

4.7 Example 7: Damped Harmonic Oscillator

Take

$$\mathcal{L} = \frac{1}{2}m\dot{x}^2 - \frac{1}{2}kx^2$$

and

$$Q^{(\text{nc})} = -c\dot{x}.$$

Then

$$\frac{d}{dt}(m\dot{x}) - (-kx) = -c\dot{x}.$$

Hence

$$m\ddot{x} + kx + c\dot{x} = 0.$$

This illustrates how a non-conservative force enters the generalized equation.

Chapter 5

A Coherent Problem-Solving Algorithm

When solving a new Lagrangian problem, follow this sequence.

1. Draw or mentally describe the system.
2. Identify constraints.
3. Count the degrees of freedom.
4. Choose independent generalized coordinates.
5. Write every required position vector in terms of those coordinates.
6. Differentiate to obtain velocities.
7. Compute the kinetic energy carefully.
8. Compute the potential energy if applicable.
9. Form $\mathcal{L} = T - V$.
10. Look for cyclic coordinates before differentiating.
11. Apply Lagrange's equation for every independent generalized coordinate q_j ,

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0.$$

12. Simplify.
13. Check dimensions, limiting cases and signs.

Most mistakes occur before differentiation. A wrong coordinate choice or wrong kinetic energy cannot be repaired by correct calculus.

Chapter 6

Useful Formula Sheet - Units I and II

$$\begin{aligned}f(q_1, \dots, q_n, t) &= 0, \\s &= 3N - k, \\\mathbf{r}_i &= \mathbf{r}_i(q_1, \dots, q_s, t), \\\mathbf{v}_i &= \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \dot{q}_j + \frac{\partial \mathbf{r}_i}{\partial t}, \\\delta \mathbf{r}_i &= \sum_j \frac{\partial \mathbf{r}_i}{\partial q_j} \delta q_j, \\\delta W &= \sum_i \mathbf{F}_i \cdot \delta \mathbf{r}_i = \sum_j Q_j \delta q_j, \\Q_j &= \sum_i \mathbf{F}_i \cdot \frac{\partial \mathbf{r}_i}{\partial q_j}, \\\sum_i (\mathbf{F}_i - m_i \mathbf{a}_i) \cdot \delta \mathbf{r}_i &= 0, \\\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_j} \right) - \frac{\partial T}{\partial q_j} &= Q_j, \\Q_j^{(c)} &= -\frac{\partial V}{\partial q_j}, \\\mathcal{L} &= T - V, \\\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} &= 0, \\p_j &= \frac{\partial \mathcal{L}}{\partial \dot{q}_j}, \\\frac{\partial \mathcal{L}}{\partial q_j} = 0 &\Rightarrow p_j = \text{constant}, \\E &= \sum_j \dot{q}_j p_j - \mathcal{L}, \\\frac{dE}{dt} &= -\frac{\partial \mathcal{L}}{\partial t}, \\\frac{\partial \mathcal{L}}{\partial t} = 0 &\Rightarrow E = \text{constant}.\end{aligned}$$

Chapter 7

Tricks, Checks and Common Mistakes

7.1 Fast Memory Tricks

- DOF means independent coordinates left after applying the constraints.
- δ means an allowed virtual variation at fixed time; d describes an actual differential change.
- Virtual work: $\delta W = \sum_j Q_j \delta q_j$.
- D'Alembert: Newton's law written in virtual-work form.
- Conservative generalized force: $Q_j = -\partial V / \partial q_j$.
- Lagrangian: $\mathcal{L} = T - V$.
- **Lagrange equation memory line:** differentiate the generalized momentum with time, then subtract the coordinate derivative:

$$\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right) - \frac{\partial \mathcal{L}}{\partial q_j} = 0$$

- Compact chain: $T \rightarrow V \rightarrow \mathcal{L} = T - V \rightarrow \partial / \partial \dot{q} \rightarrow d/dt \rightarrow \partial / \partial q$.
- Cyclic coordinate: absent from $\mathcal{L} \Rightarrow$ conjugate momentum $p_j = \partial \mathcal{L} / \partial \dot{q}_j$ is conserved.
- No explicit time dependence of $\mathcal{L} \Rightarrow$ the energy function is conserved.

7.2 Common Mistakes

1. Treating dependent coordinates as independent.
2. Confusing δq with dq .
3. Writing $\mathcal{L} = T + V$.
4. Losing the minus sign in $Q_j = -\partial V / \partial q_j$.
5. Saying “cyclic coordinate is constant” instead of “conjugate momentum is constant.”
6. Choosing inconsistent positive directions for coordinates and potential energy.
7. Ignoring explicit time dependence in coordinate transformations.
8. Using $\sin \theta \simeq \theta$ for large angles.

Chapter 8

Examination-Oriented Questions

8.1 Unit I

1. Define a constraint and classify constraints with suitable examples.
2. Explain degrees of freedom and configuration space.
3. Explain generalized coordinates and derive the transformation equation for velocity.
4. Define virtual displacement and distinguish it from actual displacement.
5. Define virtual work and generalized force.
6. State and explain the principle of virtual work.
7. Explain D'Alembert's principle and its physical meaning.
8. Starting from D'Alembert's principle, derive the generalized dynamical equation.
9. Give a complete derivation of Lagrange's equation from D'Alembert's principle.

8.2 Unit II

1. Define the Lagrangian and explain the Lagrangian method.
2. Derive Euler-Lagrange equations for a conservative system.
3. Explain non-conservative generalized forces.
4. Compare Newtonian and Lagrangian formulations.
5. Define cyclic coordinates and prove conservation of conjugate momentum.
6. Derive the energy integral and state the condition for energy conservation.
7. Explain the quadratic nature of the kinetic-energy function.
8. Derive the equation of motion of a simple pendulum using Lagrange's equation.
9. Derive the acceleration of an Atwood machine using Lagrange's equation.
10. Obtain the equations for a particle in a central potential.
11. Obtain the equation of a damped harmonic oscillator using generalized forces.

Chapter 9

References and Further Reading

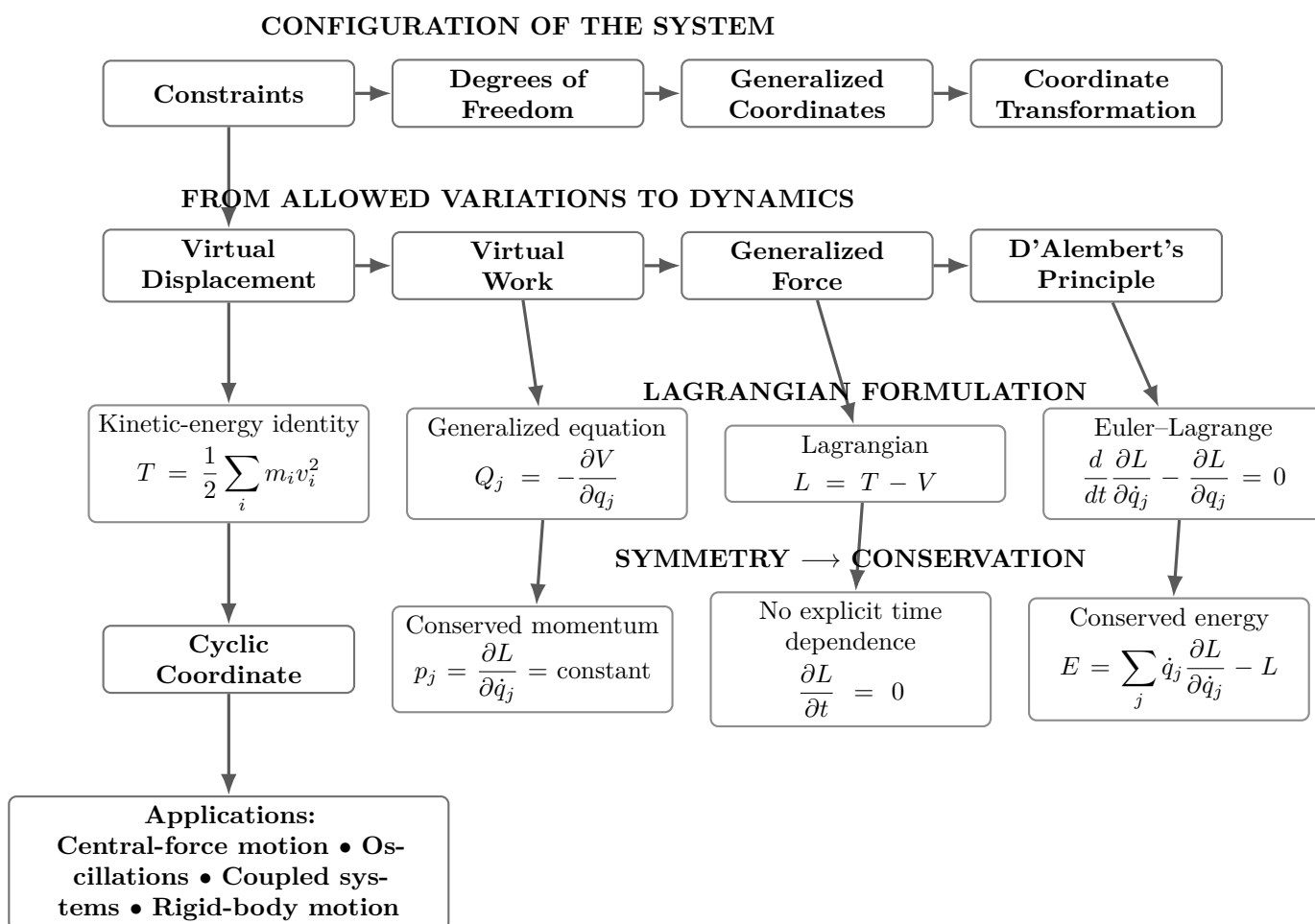
Source note. The additional derivation sequence, comparison points, cyclic-coordinate argument, and worked-example emphasis in this revision have been incorporated from the handwritten material supplied with these notes, while the overall scope follows the course syllabus.

These notes are original teaching material organized around the supplied B.Sc. Physics syllabus. Standard references may be consulted for deeper derivations and additional problems.

1. D. C. Tayal, *Mechanics*, Himalaya Publishing House, New Delhi, 2013.
2. Herbert Goldstein, Charles Poole and John Safko, *Classical Mechanics*, Pearson.
3. Marion and Thornton, *Classical Dynamics of Particles and Systems*.
4. Fowles and Cassiday, *Analytical Mechanics*.
5. Landau and Lifshitz, *Mechanics*.
6. Department syllabus supplied for the B.Sc. Physics course - used as the scope for Units I and II.

Chapter 10

Final One-Page Revision Map



If this chain is understood, Units I and II become one connected framework rather than a collection of unrelated formulas.

Figure 10.1: Final one-page revision map of analytical mechanics.